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BioHackEU26 report for Project 10: Enabling workflows and EDAM annotations to display in community website deployments

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Note: Please keep sections to a maximum of only two levels.

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Introduction

Modified and extended project abstract from BioHackathon 2026:

The ELIXIR Toolkit Theme (ETT) enables diverse research and infrastructure communities to easily deploy static websites using a GitHub repository. This is beneficial as it facilitates broad and open contributions and collaborations, provides version control and development flexibility, and streamlines the process of building and populating a website. The popularity of the lightweight and standardised ETT model is demonstrated by numerous deployments, including RDMkit (Alper et al., 2025), RSQkit (Community, 2025), and WorkflowHub documentation (Gustafsson et al., 2025).

Crucially, these websites often centralise the sharing of digital resources that can be reused by specific communities. For example, the Australian Structural Biology Computing (ASBC) Community maintains an ETT website as a shared space: the intention is for the community to be able to find activities, resources, news, and events, as well as how to join the community conversation (Michie et al., 2025). In this context, *resources* refers to any digital object that supports the requirements of the community, including, but not limited to, software tools and workflows.

Including these resources in an ETT website currently requires manual curation, which is time consuming for ETT website maintainers and does not directly support making the software and workflows findable, accessible, interoperable, and reusable (FAIR) (Barker et al., 2022, wilkinson2025applying). The ideal approach would be to use the existing FAIR infrastructure ecosystem to register software and workflows, enable the ETT to interoperate with this ecosystem, and then automatically populate a website with the relevant software or workflows. This approach leverages the registries to both curate and make resources FAIR, shares the curation burden across the contributing communities, and provides the opportunity to simultaneously add ontology annotations from registry entries to ETT websites.

The ETT already offers the possibility of annotating pages with identifiers from registries like FAIRsharing (Sansone et al., 2019) and bio.tools (Mendes et al., 2026). However, support for workflows and ontologies is not implemented. This project thus worked to extend the ETT and enable:

1. Automation of workflow imports from WorkflowHub,
2. Creation of interactive tables to present the communities workflows, and
3. Automated addition of EDAM ontology annotations at both the workflow table and webpage (i.e. schema.org) level.

The latter will build on the existing integration with bio.tools, offer the possibility of manual curation, and increase the findability of these webpages. Moreover, ontology terms (e.g. EDAM)

are already used to annotate the bio.tools elements already included on many ETT instances, including RDMkit.

Workflow developers from any community can thus continue using WorkflowHub to make their workflows FAIR, and community coordinators can independently and automatically import these to an ETT deployment along with EDAM annotations. This support would be beneficial for increasing the findability of the pages and enabling better integration across multiple instances of ETT.

Results

Project goals

1. Translate a proof of concept integration with the WorkflowHub API to the ELIXIR Toolkit Theme, including GitHub actions, configs, htmls, and css.
2. Enable support for annotation with ontology terms on ETT, including import of ontology annotations from bio.tools based on existing annotations of pages.
3. Create working examples, including documentation, that can be included on the ETT test deployment page: e.g. https://elixir-belgium.github.io/elixir-toolkit-theme/all_trainings

WorkflowHub import

Idea for table include https://australianbiocommons.github.io/workflowfinder/

Source collection on WorkflowHub: https://workflowhub.eu/collections/40

EDAM annotations

Discussion

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Author contributions

Ove Johan Ragnar Gustafsson: Conceptualization, writing – original draft.

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Appendices

If you want the Appendix (-ces) to show up after the references, wrap them in after the header, like done in this Markdown file. Look at the source to see the exact structure.